Cody / Planning Atlas

Screen journeys, a reusable character and the shared color system.

REVIEW FIRST.

PROTOTYPE NEXT.

A working atlas for the round display: first boot, essential OOBE, everyday use and future interactions.

Every screen has a writing area. Use it to review copy, touch targets, transitions and the character's response before implementation.



466 x 466

CURRENT FOCUS

Character, setup, companion status and battery notices.
Voice and OTA are explicitly future implementation work.

A4 LANDSCAPE / ENGLISH / v1

38 entries + layout variants + expressions + 256 slots

Copyright umbertogiacobbi.biz 2026 - confidential

Screen inventory

Thirty-eight entries, with pause/end and sleep/wake expanded into separate review views.

01	Sleeping
02	Waking up
03	Ready
04	Hi, I'm Cody!
05	Let's pair
06	Scan to install
07	Pairing...
08	Try again
09	You're online.
10	Make it yours
11	Cody's look
12	Connect Codex
13	You're all set.
14	Looking for your PC...
15	Connected
16	Idle
17	Project active
18	Working...
19	Response ready
20	Paused / Session ended
21	PR pending
22	Checking...
23	Checks failed
24	Failed checks
25	Retired
26	Question
27	Choose an option
28	Recording...
29	Transcribing...
30	Review transcript
31	Reply sent
32	Connection lost
33	Low battery
34	Charging
35	Sleep / Wake
36	Settings
37	Rig view
38	Expression channels

Illustrated source contracts; voice28-30 remains planned. Setup39-42 and panels are detailed later.

Copyright umbertogiacobbi.biz 2026 - confidential

The circle is usable space

Plan both the central square and the rounded regions.



CANVAS

466 x 466; center (233, 233); radius 233.

SAFE SQUARE

320 x 320 at (73, 73); bounds [73, 393).

BACK ICON

60 x 60 at (10, 203); center (40, 233).

ROUNDED REGIONS

73 px beside the square at its midline; about 9.38 px at its top and bottom.

ROTATION

Shared display/touch coordinates at 0/90/180/270 degrees. Back stays on the visual left.

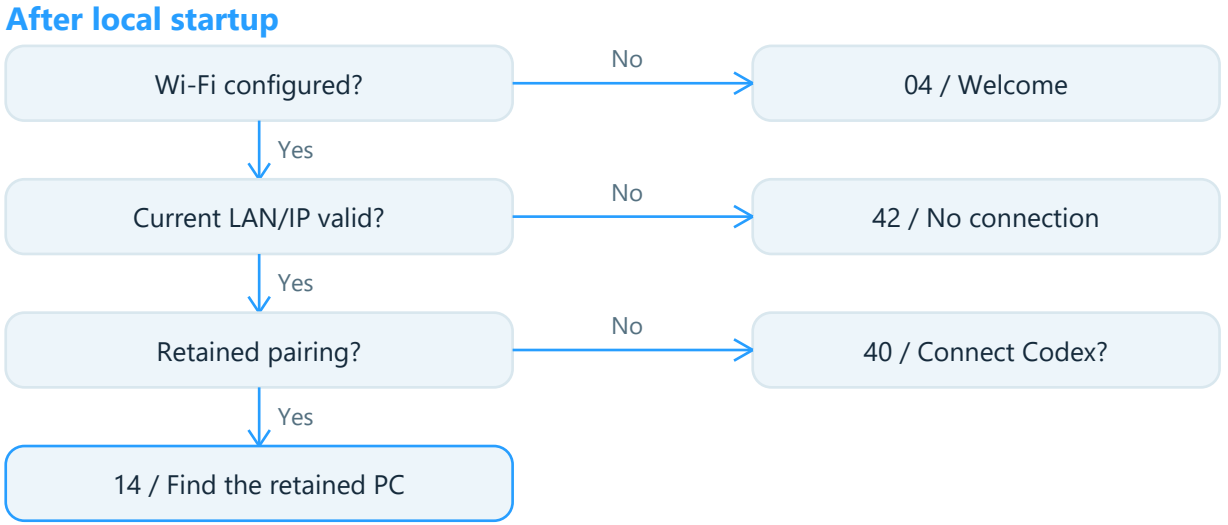
NOTES

Umberto Giacobbi | hello@umbertogiacobbi.biz
Copyright umbertogiacobbi.biz 2026 - confidential

Cody's current cold boot

Branding precedes screens 01-03; route selection uses saved state.





BOOT

Identity and timing

Cody AI Companion and A distraction-free sidekick appear together, above the complete floppy artwork. The original 42-frame animation retains its 3.5-second cycle; startup work can extend wall time.

Layout and next route

Title y=60; subtitle y=110; artwork (137,164), 192 x 219. After local startup, absent Wi-Fi opens Welcome 04; missing association/IP opens 42; a valid LAN without pairing opens 40; retained pairing searches its PC on 14.

NOTES

STATIC SCHEMATICS / illustrative fonts and values / no device capture

Meet Cody and set up Wi-Fi

Installation and Wi-Fi provisioning are separate from Codex association.

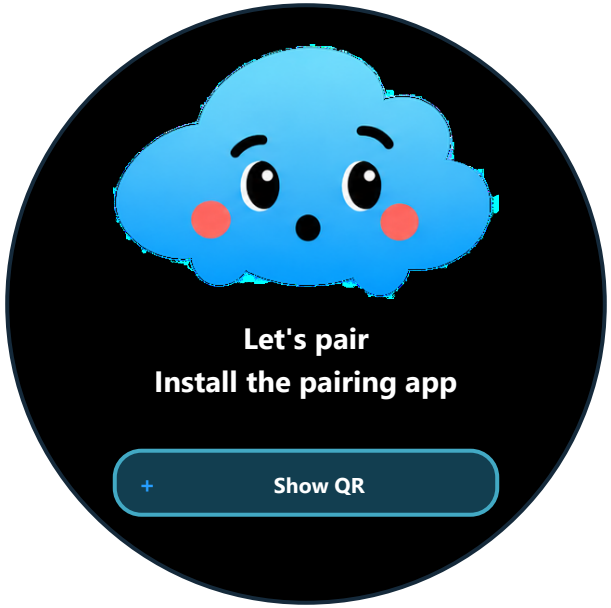
04 / Welcome



Welcome opens the local network journey. A stationary hold opens Setup when no operation is busy.

NOTES

05 / Let's pair



Show QR opens 06 without starting Bluetooth. Controls appear together after the main copy.

NOTES

06 / Install the app



Use the real installation QR with its quiet zone. This illustration is not a scannable code. Pair Cody starts provisioning 07.

NOTES

Finish network setup with real state

A current valid Wi-Fi association and IP are required before Continue.

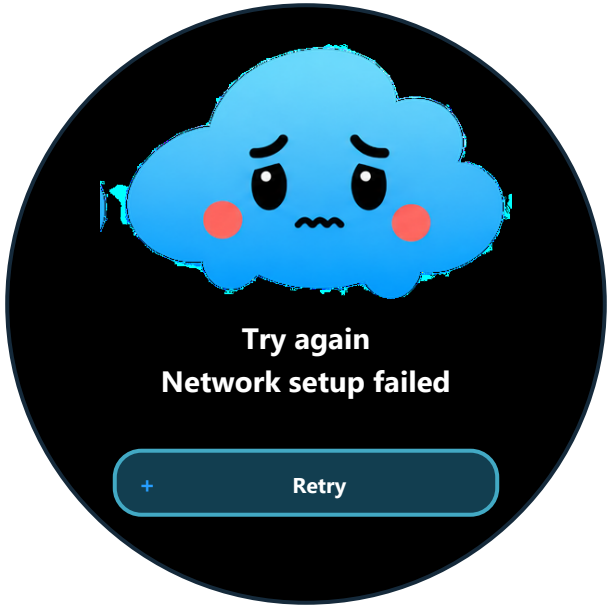
07 / Pairing



The session QR exists only while ReadyToScan. Retire it during credential transfer; Cancel waits for confirmed cleanup.

NOTES

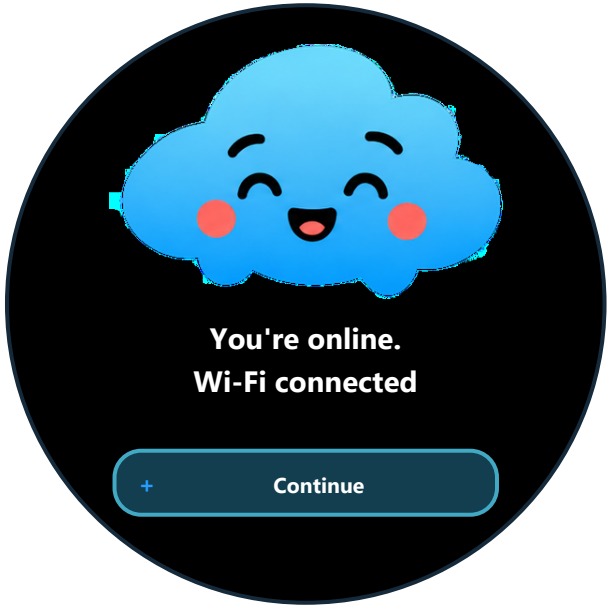
08 / Try again



Show the real failure. Retry returns to the appropriate network step; never substitute failure with online state.

NOTES

09 / You're online



Continue requires current valid Wi-Fi/IP. LAN association does not prove Internet access or Codex presence.

NOTES

Preferences and Cody's look

10 ->36 ->11; Setup Options returns to its caller and preserves network/pairing.

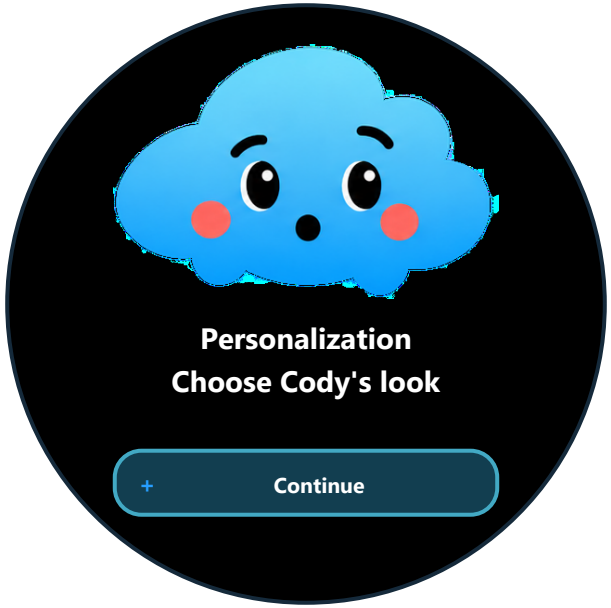
10 / Make it yours



Preferences persist only after successful storage. Sounds Off stops output; microphone and wake word continue.

NOTES

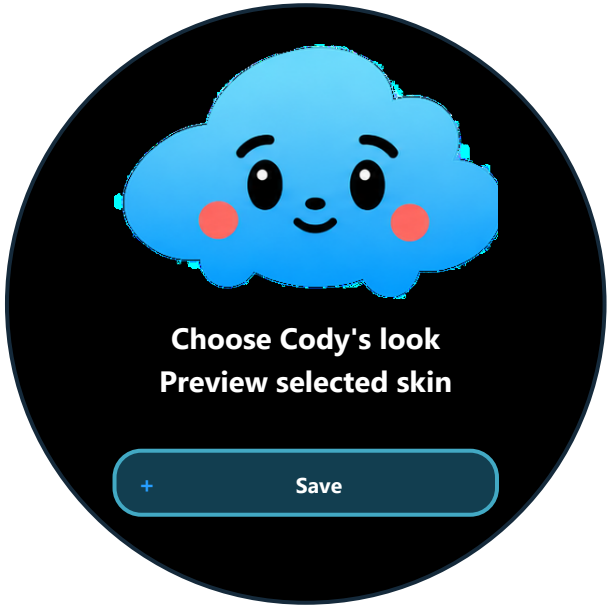
36 / Personalization



Continue opens the stable-ID skin selector. No implicit reset, silent correction or association change.

NOTES

11 / Cody's look



Pearl Mist, Aurora Veil, Cosmic Stardust, Enchanted Mosaic, Dreamwave. Save failures preserve the cause and previous stored value.

NOTES

Pair Codex, then find the retained PC

Association is durable; a live PC connection requires fresh authenticated proof.

12 / Connect Codex



About 30 seconds is an estimate. Enter all six characters on the intended Cody; association follows both persisted sides.

NOTES

13 / You're all set



Let's go immediately searches the retained PC. Successful association alone does not establish current activity.

NOTES

14 ->15 / Find the PC



Fresh authenticated presence admits Connected. Wait for a genuine prompt hook before following a chat.

NOTES

Cody follows the work you submit

16/17/18/18U; identity comes from genuine prompt hooks, not tab clicks.

16 /17 / Home and context



Home breathes and blinks. The latest genuine prompt selects context; project labels are display text, not identities.

NOTES

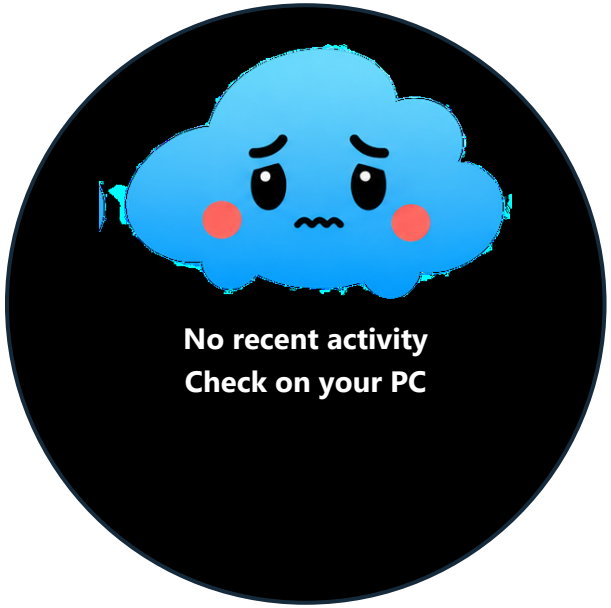
18 / Working



Tool hooks update activity for the followed chat. Events from background chats cannot steal context.

NOTES

18U / Unavailable



After 120 seconds without a new hook, current work is unavailable. An authenticated PC heartbeat does not prove ongoing work.

NOTES

Reply, pause and return to rest

Response completion, task success and link health remain separate facts.

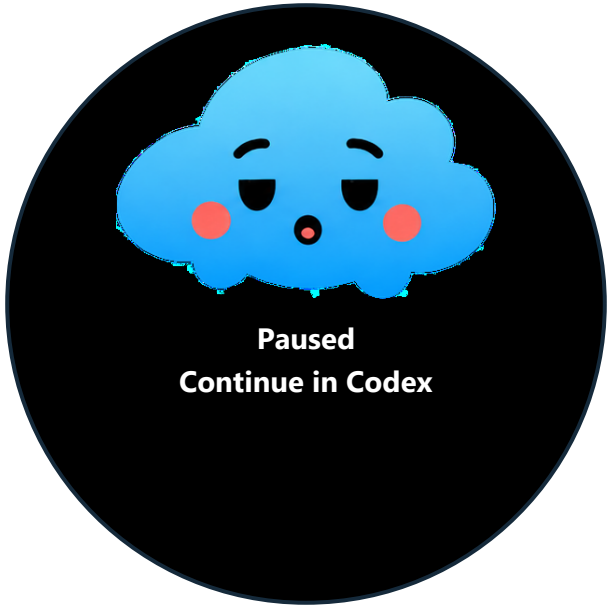
19 / Response ready



An attention cue invites reading on the PC. It does not certify tests, publication or task success.

NOTES

20A / Paused



Preserve the followed context and any open question. Routine status changes never replace a pending decision.

NOTES

20B ->16 / Ended



The character settles back to Home. A broker restart waits for a new genuine prompt rather than inventing context.

NOTES

Work status can include PR observations

21-24 are status variants; Cody does not manage pull requests or native approvals.

21 / PR pending



An actual observation can produce a concise notice in the companion workflow. No separate PR management application.

NOTES

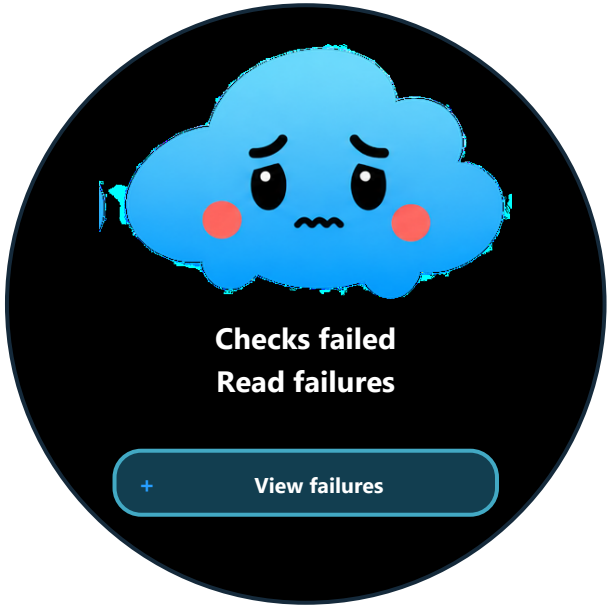
22 / Checks running



Only show reported status and measured counts when supplied. Missing data never becomes invented progress.

NOTES

23 ->24 / Checks failed



Existing View failures shows real names only, then returns to context. Logs, diagnosis and fixes remain on the PC;25 is retired.

NOTES

A question deserves its own attention

26/27; permission requests summon attention, while native approval remains on the PC.

26 / Codex needs you



An explicit MCP caller supplies English ASCII text. Caller-declared context does not prove a verified hook identity.

NOTES

27 / Keep the question



An open question is protected for 180 seconds. Battery, charging, routine notices and later activity cannot hide it.

NOTES

PC / Native approval



Cody may request attention. It never provides generic Allow/Deny based on incomplete text or replaces native approval.

NOTES

Choose, review, then explicitly Send

Two side-by-side cards; add a centered third card when needed. Stable choice IDs survive reordering.

27 / Two choices



Selecting a card opens its description; it never sends. Complete controls appear after title/body narration.

NOTES

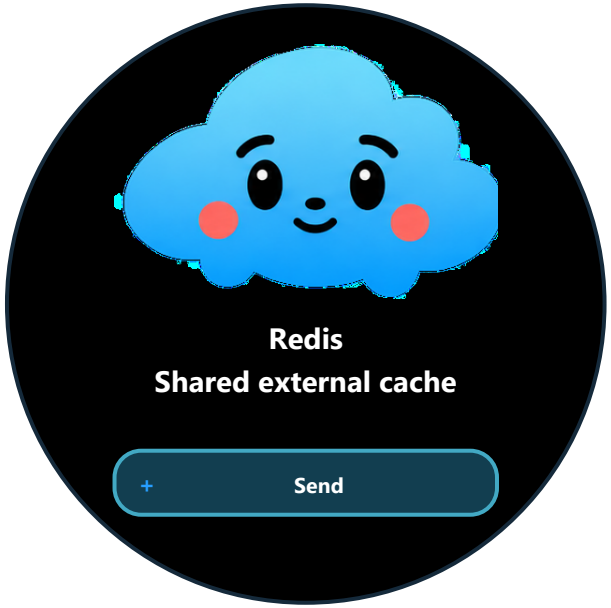
27 / Three choices



The third choice sits below the pair. Keep lateral Back separate and preserve the circle's available chord.

NOTES

27 / Review choice



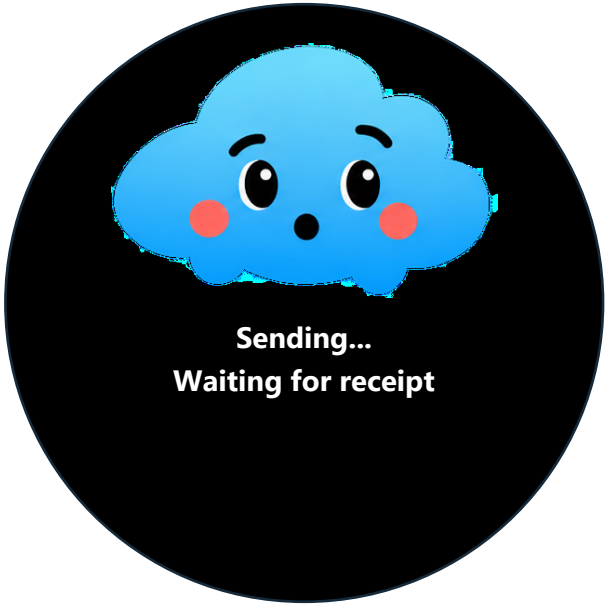
Read the selected description and press Send. Back returns to choices without silently changing the choice identity.

NOTES

A receipt is not confirmed delivery

31; received, delivered, cancellation and expiry are separate outcomes.

27 / Sending



Send fixes the selected answer. Matching repeated answers are idempotent; a different answer cannot replace it.

NOTES

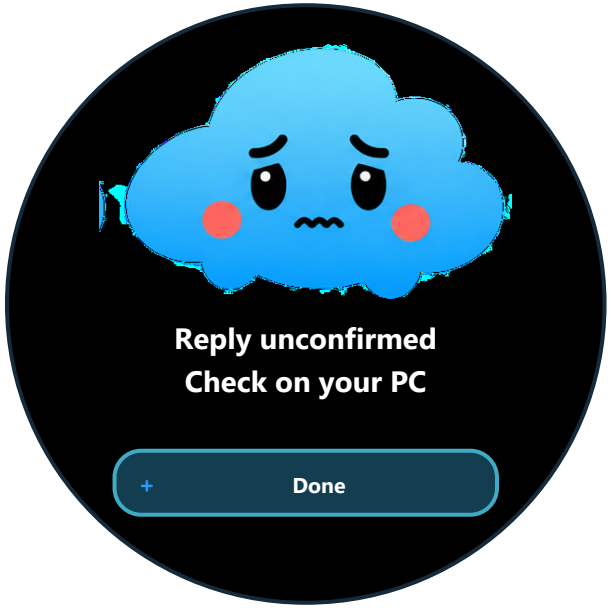
31 / Reply sent



Show only after the intended MCP caller acknowledges the matching answer_id. Admission or PC receipt alone is insufficient.

NOTES

31 / Unconfirmed



At expiry after Send, only the acknowledgement may be missing. Before Send use Request expired. Terminal requests never reopen.

NOTES

Battery notices feel like Cody

Admit after 10 seconds of uninterrupted free Home; show manga and caption together for 4 seconds.

33 / Low battery



At 20% use this notice; at 10%: Please charge me soon. Remind after 5 minutes /1 minute respectively. No percentage in the caption.

NOTES

34 / Charging starts



One notice on actual entry into charging. Ongoing charging is silent. No connected-to-power inference and no numeric overlay.

NOTES

34 / Fully charged



One Joy notice only at fresh measured 100%. Canonical valid battery data only; notices never interrupt work/questions or extend standby.

NOTES

Sleep, wake and restore the route

35A/35B and 32; sleeping is presentation, not cancellation of a pending question.

35A / Sleep



Close the eyes with a slow breath. Battery notices do not delay standby. Retain Wi-Fi on paired Cody and preserve AFE/wake word.

NOTES

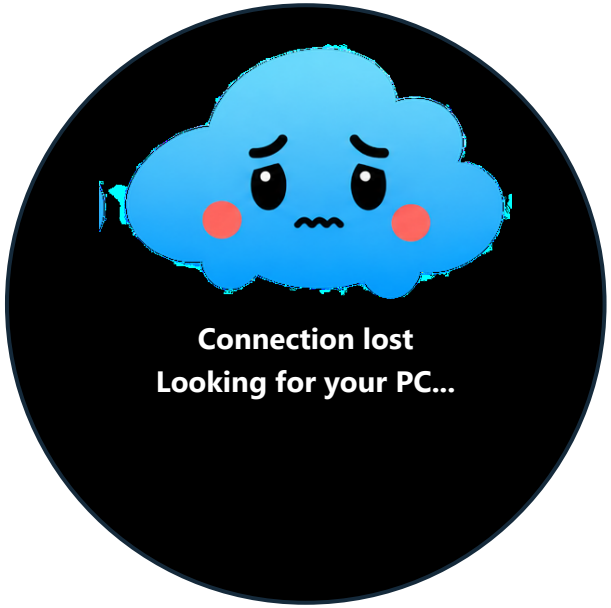
35B / Wake



Restore the prior route and selection. A touch held across wake starts its 1.5-second Setup hold only after a fresh visible acquired sample.

NOTES

32 / Link lost



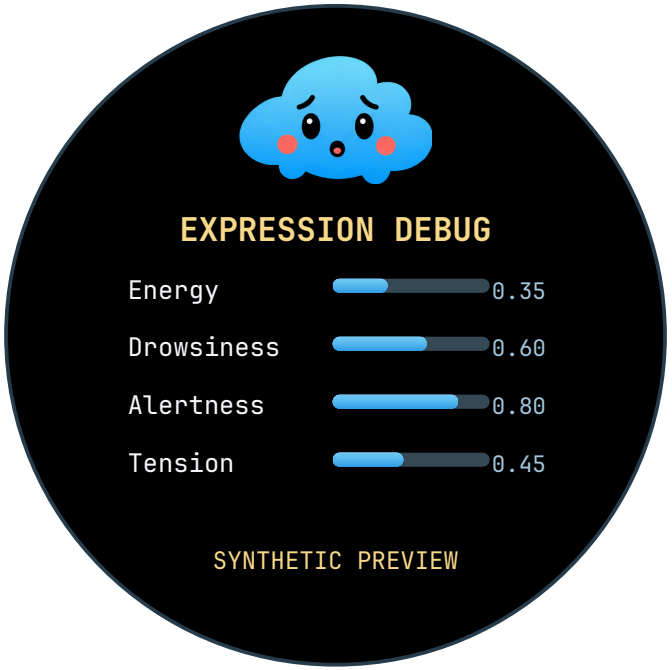
Preserve context while searching for the retained PC. Question expiry continues; disconnect and timeout never choose an answer.

NOTES

Inspect the composed expression

Debug shows the same evaluated pose that the character renderer uses.

38 / Expression channels



ONE CHARACTER, MANY CHANNELS

Inputs

Valid local events: motion, touch, orientation, charge and time since interaction.

Expressive state

Energy, drowsiness, alertness, tension and mood can coexist.

Independent motion

Breathing, gaze and blinking compose with the selected expression.

Shared geometry

Rig anchors and curve handles come from the same evaluated pose.

Preview controls

Synthetic examples are confined to a clearly labelled desktop preview.

NOTES

NOTES

Umberto Giacobbi | hello@umbertogiacobbi.biz
Copyright umbertogiacobbi.biz 2026 - confidential

Expression combinations / 01

Illustrative moments made from independent channels; these are not exclusive face states.



Calm

Relaxed brows + open eyes + gentle smile.

Slow breath; occasional blink.

NOTES



Curious

Raised brow + shifted gaze + small open mouth.

Keep the body breathing independently.

NOTES



Happy

Smiling eyes + lifted brows + open smile.

A brief response can settle back into idle.

NOTES



Focused

Attentive eyes + gathered brows + quiet mouth.

Subtle breath keeps the pose alive.

NOTES

Umberto Giacobbi | hello@umbertogiacobbi.biz
Copyright umbertogiacobbi.biz 2026 - confidential

Expression combinations / 02

Illustrative moments made from independent channels; these are not exclusive face states.



Afraid

Wide eyes + worried brows + breathing mouth.

Alertness rises quickly, then eases down.

NOTES



Stressed

Tense brows + tired eyes + a tight mouth.

Tension can coexist with low energy.

NOTES



Sleeping

Closed eyelids + relaxed brows + open mouth.

Slow breath continues with closed eyes.

NOTES



Drowsy

Half-closed eyes + relaxed brows + open mouth.

A surprise may briefly reopen the eyelids.

NOTES

Umberto Giacobbi | hello@umbertogiacobbi.biz
Copyright umbertogiacobbi.biz 2026 - confidential

The shared palette / all 256 slots

Exact RGB888 design swatches and indices from palette256.json.

000 #000000	001 #FFFFFF	002 #F4F6FA	003 #04101A	004 #0B1118	005 #111A24	006 #182330	007 #203040	008 #2A3C4D	009 #35495B	010 #40576A	011 #4D6477	012 #5B7184	013 #6B8092	014 #7D91A2	015 #92A3B2
016 #A5B4C1	017 #B9C6D1	018 #CBD6DF	019 #DCE4EA	020 #75CBF3	021 #74CAF3	022 #73CAF3	023 #72C9F3	024 #71C9F2	025 #70C8F2	026 #6FC7F2	027 #6EC7F2	028 #6DC6F2	029 #6CC6F2	030 #6BC5F1	031 #6AC4F1
032 #69C4F1	033 #68C3F1	034 #67C3F1	035 #66C2F1	036 #66C2F1	037 #65C1F0	038 #64C0F0	039 #63C0F0	040 #62BFF0	041 #61BFF0	042 #60BEF0	043 #5FBDEF	044 #5EBDEF	045 #5DBCEF	046 #5CBCEF	047 #5BBBEF
048 #5ABAEF	049 #59BAEE	050 #58B9EE	051 #57B9EE	052 #56B8EE	053 #55B7EE	054 #54B6EE	055 #53B6ED	056 #52B5ED	057 #51B4ED	058 #50B3ED	059 #4FB2EC	060 #4DB2EC	061 #4CB1EC	062 #4BB0EC	063 #4AAFEC
064 #49AEEB	065 #48AEEB	066 #47ADEB	067 #46ACEB	068 #45ABEA	069 #44AAEA	070 #43A9EA	071 #42A9EA	072 #41A8E9	073 #40A7E9	074 #3FA6E9	075 #3EA5E9	076 #3CA5E9	077 #3BA4E8	078 #3AA3E8	079 #39A2E8
080 #38A1E8	081 #37A1E7	082 #36A0E7	083 #359FE7	084 #9EF0D8	085 #9CEFD7	086 #9AEFD6	087 #97EED6	088 #95EED5	089 #93EDD4	090 #91ECD3	091 #8EECD2	092 #8CEBD2	093 #8AEBD1	094 #88EAD0	095 #85E9CF
096 #83E9CE	097 #81E8CD	098 #7FE8CD	099 #7CE7CC	100 #7AE6CB	101 #78E6CA	102 #76E5C9	103 #74E5C9	104 #71E4C8	105 #6FE3C7	106 #6DE3C6	107 #6BE2C5	108 #68E2C5	109 #66E1C4	110 #64E1C3	111 #62E0C2
112 #5FDFC1	113 #5DDFC1	114 #5BDEC0	115 #59DEBF	116 #57DDBE	117 #54DCBD	118 #52DCBD	119 #50DBBC	120 #4EDBBB	121 #4BDABA	122 #49D9B9	123 #47D9B8	124 #45D8B8	125 #42D8B7	126 #40D7B6	127 #3ED6B5
128 #3CD6B4	129 #39D5B4	130 #37D5B3	131 #35D4B2	132 #FF9385	133 #FE9183	134 #FE8E81	135 #FD8C7F	136 #FC8A7D	137 #FC877B	138 #FB8579	139 #FA8377	140 #FA8075	141 #F97E73	142 #F87C71	143 #F8796F
144 #F7776D	145 #F7746B	146 #F67269	147 #F57067	148 #F56D65	149 #F46B63	150 #F36961	151 #F3665F	152 #F2645D	153 #F1625B	154 #F15F59	155 #F05D57	156 #F8D989	157 #F6D685	158 #F5D281	159 #F3CF7D
160 #F1CC79	161 #F0C875	162 #EEC571	163 #ECC26D	164 #EBBE68	165 #E9BB64	166 #E7B860	167 #E6B45C	168 #E4B158	169 #E2AE54	170 #E1AA50	171 #DFA74C	172 R #000000	173 R #000000	174 R #000000	175 R #000000
176 R #000000	177 R #000000	178 R #000000	179 R #000000	180 R #000000	181 R #000000	182 R #000000	183 R #000000	184 R #000000	185 R #000000	186 R #000000	187 R #000000	188 R #000000	189 R #000000	190 R #000000	191 R #000000
192 R #000000	193 R #000000	194 R #000000	195 R #000000	196 R #000000	197 R #000000	198 R #000000	199 R #000000	200 R #000000	201 R #000000	202 R #000000	203 R #000000	204 R #000000	205 R #000000	206 R #000000	207 R #000000
208 R #000000	209 R #000000	210 R #000000	211 R #000000	212 R #000000	213 R #000000	214 R #000000	215 R #000000	216 R #000000	217 R #000000	218 R #000000	219 R #000000	220 R #000000	221 R #000000	222 R #000000	223 R #000000
224 R #000000	225 R #000000	226 R #000000	227 R #000000	228 R #000000	229 R #000000	230 R #000000	231 R #000000	232 R #000000	233 R #000000	234 R #000000	235 R #000000	236 R #000000	237 R #000000	238 R #000000	239 R #000000
240 R #000000	241 R #000000	242 R #000000	243 R #000000	244 R #000000	245 R #000000	246 R #000000	247 R #000000	248 R #000000	249 R #000000	250 R #000000	251 R #000000	252 R #000000	253 R #000000	254 R #000000	255 R #000000

172 allocated slots / 84 reserved black slots / R = reserved, not an extra black role

Design colors are sRGB. Several entries may map to the same RGB565 word; 256 slots do not imply 256 unique colors.

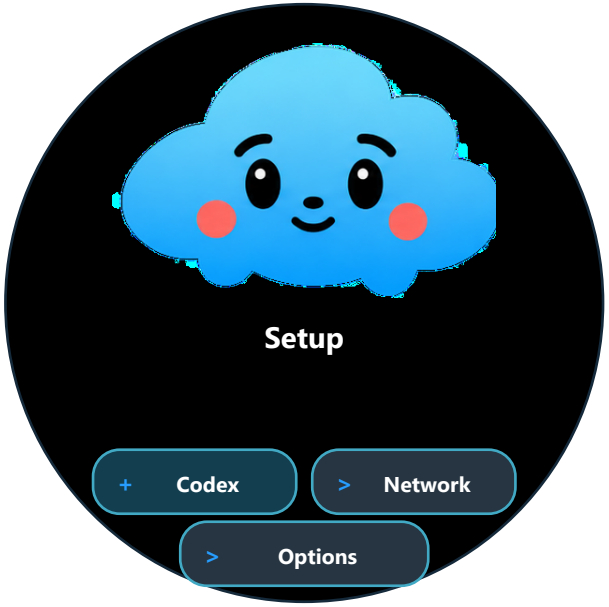
NOTES

Copyright umbertogiacobbi.biz 2026 - confidential

Setup keeps intent and routes explicit

39-42; local Wi-Fi setup and Codex association have independent owners and purposes.

41 / Setup



Hold the visible character still for 1.5 seconds; release before selecting. Keep valid previous-route lateral Back separate.

NOTES

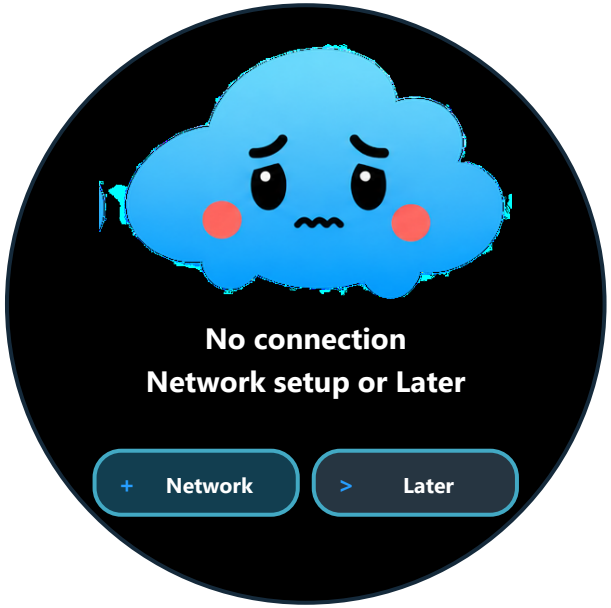
40 / Retained pairing



Keep searches the saved PC; Pair again explicitly admits replacement. Forget requires a separate confirmation and preserves unrelated settings.

NOTES

42 /39 / Network and handoff



No valid association/IP ->42. Debug handoff 39 requires a fresh tap; Release continues to 09. No public Internet requirement.

NOTES

Technical panels, then back to Cody

Passive panel taps cycle; a deliberate shake returns directly to Home. These panels have no Back.

PANEL / Activity



A short passive tap advances to Model & Context. Retain real validity, timestamps and failure state; no fabricated metrics.

NOTES

PANEL / Model & Context



A short passive tap advances to Session & Link. Model and context are reported fields, not inferred from the visible character.

NOTES

PANEL / Session & Link



A short passive tap returns to Activity. Deliberate shake returns Home. Setup and remote questions retain their own routes.

NOTES

Planned voice: wake, capture, send

28/29; one-way voice to Codex is a design proposal, not implemented firmware or a verified service.

28 / Start voice



Wake word, double tap or button requests the same priority event. Do not start a second capture. Explicit request/approval behavior needs design before implementation.

NOTES

28 / Capture to Wi-Fi



Suspend character rendering. Use a small precomputed animation; this static glyph is illustrative, not a measured waveform.

NOTES

29 / Processing



A dedicated authenticated encrypted channel streams canonical audio acquisition to the PC. Native Codex audio support requires installed-version/account verification.

NOTES

Planned voice: bounded buffer and outcomes

30/31; ten seconds is a transport-stall budget, not the intended maximum utterance duration.

28 / PC slows down



Proposed PSRAM queue: up to 10 seconds of unacknowledged PCM16 mono 16kHz,320000 bytes plus metadata. Stop explicitly at capacity; never overwrite unheard words.

NOTES

30 / Review transcript



Historical review screen 30 is retained as an optional future mode. Initial wake-word dictation may use automatic end-of-speech delivery; no firmware behavior is claimed.

NOTES

31 / Delivered or failed



Received by the PC, accepted by Codex and work completed are distinct. No full duplex, SD writes or flash recording required by this proposal.

NOTES

Revision register

Historical dates, original screen references and evidence are retained in this final grid.

DATE / TYPE	REVISION DETAILS AND ORIGINAL REFERENCES
03 October 2026 ADDENDUM	**Separate app installation from pairing** Affected: Original planning page 23; original screen IDs retained in the description. Umberto Giacobbi (project owner) 05 / 06 / 21 23 / keep the existing printout 06 / REVISED 06-bis / NEW Pair Cody Scan to pair Scan inside the provisioning app. Installation QR. Remove the visible 'Scan to install' title. Pair Cody opens the device QR. Device QR. Scan it inside the provisioning app. Back stops the active BLE session and returns to 06. WHY THIS CHANGE The installation link downloads the app. A second QR identifies Cody for Wi-Fi provisioning over BLE. One QR cannot express both steps clearly. UPDATED JOURNEY 05 Show QR -> 06 06 Pair Cody -> 06-bis Credentials received -> 07 Failure -> 08 / Retry -> 06-bis Wi-Fi connected + IP -> 09 SHARED ACTION BUTTON Reuse one capsule control: Blue / Primary, Mint / Success, Gold / Warning, Coral / Danger, Neutrals / Secondary. Use the palette roles on base page 21; Retry stays Primary. QR drawings are illustrative. Firmware must supply the real app URL and device-session payload. 06-bis is a screen ID, not a PDF page number. This addendum records the owner's requested design change.
04 October 2026 ADDENDUM	**A02 / Screen 11 becomes Cody's look** Affected: Original planning page 24; original screen IDs retained in the description. Cody's look Pearl Mist Aurora Veil Cosmic Stardust Done SCREEN 11 / REVISED Affected historical pages: 02 inventory and 07 OOBE layouts. The old Blue/Mint color chips are superseded; their printed pages remain historical. Five authored looks Pearl Mist cody.skin.pearl_mist.v 1 DEFAULT Aurora Veil cody.skin.aurora_veil.v 1 Cosmic Stardust cody.skin.cosmic_stardust.v 1 Enchanted Mosaic cody.skin.enchanted_mosaic.v 1 Dreamwave cody.skin.dreamwave.v 1 Interaction contract Drag vertically to scroll. Tap one row to select exactly one look. The preview changes immediately. The user sees the cool name; firmware stores only the stable internal ID. Back or Done returns to the caller. No color editor, mixing or texture controls are added.
04 October 2026 ADDENDUM	**A03 / Pull requests become status-only** Affected: Original planning page 25; original screen IDs retained in the description. 21 / PR pending NOVA feature/cache PR #83 Pending 22 / Checks running NOVA feature/cache Checks running 3 of 6 complete 23 / Checks failed NOVA feature/cache 2 checks failed View failures Observe only. No PR action is offered. Show a count only when the exact total is known. One explicit action opens the failure list. 24 / Failed checks 2 checks failed CacheTests.Write Failed lint/cache.ts Failed Review details on your PC READ-ONLY CONTRACT Screen 22 briefly changes to 'All checks passed' before returning to screen 17. Screen 24 lists real failed check names and uses the fixed lateral Back control. It has no View log button and no selectable log rows. 25 / RETIRED Raw logs and stack traces stay on the PC. Cody does not merge, approve, close, rerun or comment on a PR.

Revision register

Historical dates, original screen references and evidence are retained in this final grid.

DATE / TYPE	REVISION DETAILS AND ORIGINAL REFERENCES
04 October 2026 ADDENDUM	**A04 / Pairing handoff and continued setup** Affected: Original planning page 26; original screen IDs retained in the description. 39 / Pairing debug DEBUG ONLY DEBUG ONLY 1. Wi-Fi connected to `Biagio` 2. IP obtained: `192.168.1.28` 3. `STA Got IP` 4. `Provisioning stopped` 5. Entered `online` screen 6. BLE restarted as `CODY-42DA` Tap anywhere to continue PAIRING SUCCESS CONTRACT A healthy current Wi-Fi/IP handoff writes cody_firstboot=false and reads it back before any success UI. Debug builds then show screen 39. One awake tap dismisses it. Release builds contain no screen-39 route and go directly to 09. SCREENS 04-09 / CONFIG_LUNASHELL_CODY_FORCE_FIRST_BOOT=n Accepts y/n; Cody profile default: n. y replays first setup at each boot, even when cody_firstboot=false; n follows the saved value. The override does not change flash. SCREEN 11 / REVISED INTERACTION Use the five-look selector shown on planning page 24. Remove Back. Pearl Mist remains visibly selected by default, but Done starts disabled. The user must tap one row explicitly; tapping Pearl Mist itself is a valid confirmation. The selector remains reusable from screen 36 later. FIRST-RUN ORDER 09 You're online 10 Make it yours 36 Settings 11 Cody's look 12 Connect Codex 13 All set 14 Looking for PC 15 Connected SCREEN 12 / CONFIG_LUNA_SHELL_CODY_ENABLE_CODEX=y
09 October 2026 CHANGE	**Startup: two saved decisions** Affected: Original planning page 27; original screen IDs retained in the description. 1 / No saved Wi-Fi profile A valid absent-profile observation opens 04 > 05 > 06 > 07 > 09. Debug may show 39 before 09. Continue: 10 > 36 > 11 (skin), then the Codex decision. 2 / Saved Wi-Fi; waiting for the network 42 / No connection keeps the profile while the Wi-Fi owner reconnects. Network setup reopens 04-09; Later returns to Idle. Invalid/stale data shows UNAVAILABLE. 3 / LAN ready; no saved Codex pairing 40 / Connect Codex? Yes starts the code on 12. No returns to 16 / Idle without a settings write or CCP request. The next boot may ask again. 4 / LAN ready; saved Codex pairing Adopt the saved attempt and search on 14. Only authenticated READY enters 15. A retained identity alone never means the PC is connected. Connect Codex? Yes No 40 / Explicit Yes or No Illustrative UI. Choice controls do not start pairing until the user acts.
09 October 2026 CHANGE	**Reopen setup from Cody** Affected: Original planning page 28; original screen IDs retained in the description. 41 / Setup Connect Codex opens a fresh display-code attempt. Network setup reopens 04-09. Left Back returns to Idle. Opening or leaving the menu erases no Wi-Fi, skin or pairing. Keep the gesture deliberate Available from stable Idle, PC search/connected and terminal CCP errors. Moving more than 12 px, a sample gap over 250 ms, rotation or a lifecycle interruption cancels the hold. Busy work and active provisioning are excluded; release before choosing. PWR / existing policy Single: standby/wake; active Debug defaults to screenshot countdown. Long release: standby. Debug double within 500 ms, or a continuous 3-second hold: restart preserving data. BOOT / existing policy Short: restore a dim/sleeping panel first, otherwise toggle fullscreen/status arcs. Hold 3 seconds: Debug restart; Release has no firmware reset action. Saved Wi-Fi and CCP survive restart. Setup Connect Codex Network setup 41 / Reversible setup access Illustrative UI. A true factory erase is a separate explicitly requested operation.

Revision register

Historical dates, original screen references and evidence are retained in this final grid.

DATE / TYPE	REVISION DETAILS AND ORIGINAL REFERENCES
09 October 2026 ERRATA	**Errata and corrections** Affected: Original planning page 29; original screen IDs retained in the description. Original analysis gap The initial analysis omitted startup with Wi-Fi already configured but no saved Codex pairing, and did not define a reversible user entry to setup. This was an original analysis gap (original analysis gap), now addressed in source. Affected original pages and screen IDs Original pages 02 (inventory), 07 (preferences/PC setup) and 26 (A04 handoff); screens 04-16, 35B, 36. The inventory gains 40, 41 and 42. Corrections Add 40 / optional Yes-No Codex choice, 41 / character-hold Setup, and 42 / reconnect wait. 09 means local Wi-Fi/IP; 11 selects a skin. No/Later performs no persistence or new CCP request. Saved pairing returns to 14 and requires authenticated READY for 15. PWR/BOOT do not clear pairing. Preservation All 26 previous pages, earlier corrections, bookmarks and unrelated content are retained. This dated addendum supersedes only the cited startup/setup/install statements; the original numbering stays stable. Evidence and limits Firmware source implementation and static integration review completed. PDF pages rendered and visually reviewed. No firmware tests, validation scans, configure/build/size, simulator execution, device pairing/input, flash or erase accompanied this revision. Runtime validation is pending.
09 October 2026 CHANGE	**Setup after standby** Affected: Original planning page 30; original screen IDs retained in the description. 04-06 / Escape the network loop After inactivity, Welcome previously sent its first touch straight back to Let's pair. Welcome now opens 41 on a stationary tap release. A character hold also opens 41 from timeout Welcome, Let's pair or the installation QR page when no operation is busy. 35B / Wake, then hold A finger already down during wake starts its 1.5-second hold only on a fresh acquired sample with the restored visible character. Hidden standby/wake time never counts. Movement over 12 px, report gaps over 250 ms or rotation cancel the hold. Release before choosing. 41 / Two explicit destinations Upper Connect Codex opens 12 and its six-character CCP code using the local Wi-Fi/IP connection. Lower Network setup explicitly opens 04-09. No Internet probe is required. Active provisioning, worker admission and pending cancellation block menu access. Setup Connect Codex Network setup 41 / Same choices after wake Illustrative UI. Existing credentials and retained CCP pairing are preserved.

Revision register

Historical dates, original screen references and evidence are retained in this final grid.

DATE / TYPE	REVISION DETAILS AND ORIGINAL REFERENCES
09 October 2026 ERRATA	**Errata and corrections** Affected: Original planning page 31; original screen IDs retained in the description. Original analysis gap The first analysis did not reconcile the legacy Welcome timeout/touch path with reopening setup after standby. This was an original analysis gap (original analysis gap). Affected pages and screens Previous setup addendum: page 28. Affected screen IDs: 04, 05, 06, 12, 35B and 41. Earlier dated errata remain intact. Corrections Timeout Welcome tap release and stationary character holds on resting 04-06 reopen 41. A hold spanning wake starts only after visible restoration. Connect Codex goes to 12; Network setup goes to 04-09. Busy work and pending cleanup cannot be bypassed. Evidence and limits A user-authorized 90-second passive serial capture confirmed LAN/DHCP, Welcome inactivity return and standby. No Connect Codex press occurred in that capture; the initial wrong-selection report remains unresolved. The correction has static source review; corrected physical behavior still needs observation. Preservation All 29 preceding pages, prior errata and unrelated content are retained. This addendum supersedes only the cited escape/gesture behavior. Saved credentials are not erased.
09 October 2026 CHANGE	**Cody action language** Affected: Original planning page 32; original screen IDs retained in the description. Layout rule Two actions: rounded cards side by side. Three: the same pair plus one centered action below. One: a wide centered capsule with its icon left of the label. Shared visual language Dark restrained gradients, thin semantic borders, white labels and action icons. Teal: affirmative/Codex; blue: network; graphite: No/Later/Options; amber: Retry/Cancel. Circular geometry Keep Back separate at the left. Share rendering and touch bounds in every orientation. QR pages keep the quiet zone and use the compact single-action variant.
09 October 2026 CHANGE	**Options and touch continuity** Affected: Original planning page 33; original screen IDs retained in the description. 41 / Setup Left card: Connect Codex. Right card: Network setup. Centered third action: Options. Options opens 10 / Make it yours, then 36 and 11 / skin. Done returns to Setup; Back retraces this local preferences path. No network or Codex association is erased or implicitly started. 40 / Connect Codex? and 42 / No connection Yes/No and Network setup/Later are side by side, with purpose-specific icons and colors. Their existing explicit backend admission stays intact. Screen IDs 05-15 and 36 share the action styling where an action is present. Observed freeze / original cause An authorized 90-second reset/capture recorded Yes at 16497 ms, entry to Connect Codex at 16498 ms, then a CST9217 report with fifteen FF bytes and ACK=FF at 17112 ms (expected AB). ESP_ERR_INVALID_RESPONSE stopped the renderer. LAN/IP was ready. Scoped runtime correction The touch owner still rejects malformed reports and publishes the real error. After startup only, that acquisition error is optional for rendering. Discard queued gesture reports, cancel the gesture and require a real acquired release before another press. No release is invented. The same acquisition error can mean an I2C NACK; its origin is not inferred. Startup, other acquisition errors, locks and route failures remain terminal. Evidence boundary Native source integration and approved mockups are documented here. The underlying cause of the FF report and corrected physical touch/standby behavior remain unverified. A successful build/flash alone does not establish those outcomes.

Revision register

Historical dates, original screen references and evidence are retained in this final grid.

DATE / TYPE	REVISION DETAILS AND ORIGINAL REFERENCES
09 October 2026 ERRATA	**Errata and corrections** Affected: Original planning page 34; original screen IDs retained in the description. Important design decision The user approved the two-column pair plus centered third action, and the wide single action with icon to the left. These are binding layouts from this revision onward. Gradients, icons and semantic colors are required across Cody screens. Affected original material Original pages 02 (inventory), 07 (preferences/PC setup) and 26 (A04 handoff); screens 04-16, 35B, 36. The inventory gains 40, 41 and 42. Also affected: screens 05-15, 35B, 36 and 40-42. Earlier stacked-blue control illustrations are superseded only in their action layout/styling. Original analysis gaps The initial analysis omitted Options re-entry and did not distinguish an optional malformed runtime touch report from renderer failure. This was a original analysis gap. Options now opens 10 and returns to Setup after preferences/skin. Corrections and limits Yes/No become horizontal and semantic. The measured invalid touch report retains its original error while rendering continues with interrupted gestures. Hardware FF root cause and physical corrected behavior remain open. Preservation All 31 preceding pages, earlier errata and unrelated content remain unchanged. Approved images are design references, not physical device screenshots.
10 October 2026 ADDENDUM	**Addendum / automatic context** Affected: Original planning page 35; original screen IDs retained in the description. 15 / Connected, waiting for Codex An authenticated PC reply proves presence. Until a genuine prompt arrives, show Waiting for Codex. There is no device picker and no manual selection command in Codex. The user prompt chooses context Trusted UserPromptSubmit sends the exact chat and turn identities to CCP. The latest prompt-hook emission becomes the followed context for this PC's paired Cody devices. SessionStart alone does not select a chat. Background chats stay in the background Tool activity, completion and interrupts from another chat cannot steal the followed context. A new prompt in that other chat switches it. A tab click alone has no documented hook signal and is not claimed as focus tracking. 32 / Recover the same context LAN reconnect restores the broker's current context. A broker restart waits for the next genuine prompt. The workspace label is display text, not identity. No transcripts, prompts or responses are retained by CCP.

Revision register

Historical dates, original screen references and evidence are retained in this final grid.

DATE / TYPE	REVISION DETAILS AND ORIGINAL REFERENCES
10 October 2026 ERRATA	**Errata and corrections** Affected: Original planning page 36; original screen IDs retained in the description. Affected original material Original pages 9-10 and 25; screens 17-20, 21-24 and 32. Analysis gap corrected Screen 15 originally proved PC presence only. The initial analysis omitted explicit chat selection, shared runtime lifetime and event freshness. This was a original analysis gap; it is now addressed in source. Contract and UI CCP 0.3 adds metadata-only hooks, exact session selection and nonce-bound ACTIVITY messages. The canonical snapshot owns validity and errors. 18U adds the unavailable state. Setup re-entry preserves the selected context. Evidence and limits The user confirmed Connected on 10 October after real pairing. That evidence belongs to the previous firmware and CCP 0.2. New code is implemented; host checks, hook trust/autoload and physical 0.3 behavior are not established here. A response-ready event is never reported as task success. Preservation All 34 previous pages and earlier errata remain unchanged. The approved gradient/icon action layouts remain binding. These diagrams describe architecture, not captured device pixels.
10 October 2026 ADDENDUM	**Addendum / current integration contract** Affected: Original planning page 37; original screen IDs retained in the description. PC ownership One local broker owns the paired PC identity and authenticated activity. Multiple MCP clients share it. Trusted hook metadata supplies session/turn/tool identities; prompts, replies, transcripts and tool arguments are discarded at ingestion. Device ownership Cody receives nonce-bound ACTIVITY alongside authenticated READY. The canonical snapshot owns validity, timestamps and real errors. UI and audio consume it; neither chooses a chat nor reads host data independently. Limits and observation Eight seconds without packets expires device activity; 120 seconds without an active hook marks activity unavailable. Missing input never becomes success. Hook acceptance is not evidence of displayed pixels or audible playback. Deployment and references CCP 0.4 removes manual select/clear tools. Install the updated plugin, review/trust its hooks and submit a real prompt. Source: learn.chatgpt.com/docs/hooks and developers.openai.com/plugins/build/plugins. Actual hook/display delivery remains to verify.

Revision register

Historical dates, original screen references and evidence are retained in this final grid.

DATE / TYPE	REVISION DETAILS AND ORIGINAL REFERENCES
10 October 2026 ERRATA	**Errata and corrections** Affected: Original planning page 38; original screen IDs retained in the description. Main pages updated Current flow is integrated from the cover onward, using screen IDs as the reference. Screens 10-15, 36, 40 and 41 now show Skin, real pairing and the agreed setup access. Screens 15-20, 18U and 32 describe automatic context and real activity. Manual selection removed The old Choose/Pick a chat prerequisite was an initial analysis gap (original analysis gap). CCP associates the latest genuine prompt automatically. Cody only presents the supplied context. Background chat events do not change it. History retained Earlier addenda, unrelated pages and errata remain at the end. The activity addendum itself is corrected. Historical errata mentioning explicit selection describe the previous decision and are superseded by this dated revision, not current instructions. Evidence and limits CCP 0.4 and the Waiting for Codex caption are source changes. New host checks, plugin hook execution and physical behavior require their own evidence. The prior successful Connected/standby capture does not verify this revision. Screen artwork is illustrative.
10 October 2026 ADDENDUM	**Addendum / retained pairing** Affected: Original planning page 39; original screen IDs retained in the description. First boot / Configured device Without saved Wi-Fi, open 04-09. With credentials but no IP, use 42. With LAN and no pairing, offer 40. A valid retained record resumes 14-15 without checking Internet access. Setup / Consent to replacement Hold the character for 1.5 seconds to open 41. A retained pairing shows Already paired: Keep pairing, Pair again or Forget. Retry resumes discovery without a new code. Pairing / Presence / Activity Screen 13 confirms pairing. Let's go immediately searches for the PC. Screen 15 confirms CCP and waits for a genuine prompt. Hooks follow its chat automatically; manual selection is removed. Options / Local choices Options opens Make it yours and reuses Settings and Skin, then returns to the correct caller. Preferences, PC replacement and explicit erasure remain separate decisions.
10 October 2026 ADDENDUM	**Addendum / recovery cases** Affected: Original planning page 40; original screen IDs retained in the description. Internet down, LAN working Continue using local Wi-Fi/IP between Cody and CCP. Codex online-service availability is separate: a PC present on the LAN cannot guarantee a model reply. Wi-Fi or local IP absent Preserve credentials and the paired PC. At startup, show 42 with Network setup or Later. Wait for LAN recovery and retry with the retained identity. Real transport errors remain visible in state. PC or CCP absent Continue searching on 14; use 32 for loss after followed activity. Retry resumes search without a new code. Holding Cody for 1.5 seconds still opens Setup. 35A / 35B and firmware update Ordinary standby preserves route and pairing. Wake restores authenticated presence. A normal application flash preserves NVS; erasure, corrupt data and a different PC identity are separate cases.

Revision register

Historical dates, original screen references and evidence are retained in this final grid.

DATE / TYPE	REVISION DETAILS AND ORIGINAL REFERENCES
10 October 2026 ADDENDUM	**Addendum / confirmation and reset** Affected: Original planning page 41; original screen IDs retained in the description. Two candidates, active records retained The PC saves its DPAPI candidate before PAIR; Cody uses ccp_pending. Previous active records remain separate. CONFIRM or fresh READY proves the PC and allows commit and readback on Cody. Close with proof of commit The PC promotes only after authenticated ACTIVE or COMMITTED, never FIND alone. After each valid READY, Cody repeats COMMITTED with the current nonce, recovering restarts and lost ACKs without another code. Cancel / End the attempt From the ready code, discard Cody's candidate and preserve the previous live link. The PC candidate remains pending. Replacement requires a new code and explicit replace_pending:true. Proof with the old key never authorizes deletion. Forget / Codex pairing only Confirm Forget pairing with Go back / Forget and Only Codex is reset. Delete only Cody's CCP records; the PC keeps its record. After the actual result, show Unpaired / Pairing removed, then Done -> Setup.
10 October 2026 ERRATA	**Errata and corrections** Affected: Original planning page 42; original screen IDs retained in the description. Original pages 1, 2, 4, 5, 6, 7, 8, 9, 10, 14, 16, 21, 22 The current flow is integrated into the main pages, preserving unrelated pages and every earlier addendum/correction. Screen IDs identify routes even when pagination changes. Screens 10-16, 32, 35B and 40-42 Options saves four preferences; Already paired offers Keep pairing, Pair again and Forget. Retry, Cancel and reset are separate. Separate journals and COMMITTED prove actual commit. Let's go searches immediately. Initial analysis gap and responsibility Commissioning, recovery and activity had been mixed without consent to replacement. Those decisions are now explicit. CCP follows genuine prompts; the old manual chat selector is retired. UI decision and evidence Preserve the approved language: gradients, icons, horizontal pairs or a single capsule, and buttons without teletype. This revision updates sources and documents; device, hook and simulator checks still pending are listed separately.
10 October 2026 ADDENDUM	**Addendum / notices and decisions** Affected: Original planning page 43; original screen IDs retained in the description. 27 / Choose a reply 27 / Review choice 26 / Codex needs you and Open Show caller-declared context, title and notice. Each request has a separate ticket and lasts 180 seconds. Opening or closing a view never extends expiry or selects an answer. Use two side-by-side cards; add a centered third card below. Each choice has a stable ID and meaning. Back returns to 26 and retains the question without sending. Selecting a choice opens a description summary with a Send capsule. Back returns to options. After Send, keep the choice fixed and show Sending reply. Never add a fourth button beneath three cards. 31 / Reply sent after delivery A received UDP ACK leaves Waiting for Codex: CCP recorded the choice. Only delivered shows Reply sent and Delivered to Codex. Done restores the prior context on 17, 18 or 16.

Revision register

Historical dates, original screen references and evidence are retained in this final grid.

DATE / TYPE	REVISION DETAILS AND ORIGINAL REFERENCES
10 October 2026 ADDENDUM	**Addendum / read-only PRs** Affected: Original planning page 44; original screen IDs retained in the description. 21 / PR pending 23 -> 24 / View failures Observe only the specified PR. Keep the exact head of the explicit snapshot from Codex. Stop does not prove check results; never offer merge, approve, close, rerun or comment. 22 / Checks running and Checks passed Counts require a complete, current list for the same head. Checks passed has no buttons: wait for narration and three seconds, then return to work. The only action opens real failed-check names: at most six, each up to 48 ASCII characters. Unrepresentable input is explicitly rejected; names are never cut and totals are never invented. 24 read-only / 25 retired Show three names per page, each up to two lines; scroll vertically and retain lateral Back. No View log or selectable rows. Logs, diagnosis and fixes stay on the PC.
10 October 2026 ADDENDUM	**Addendum / delivery and commands** Affected: Original planning page 45; original screen IDs retained in the description. received is not delivered Polling returns the choice and answer_id. A later poll with the matching acknowledge_answer_id proves caller receipt. Only then enter delivered and screen 31. Expiry and uncertain delivery The 180-second limit also applies in received. After Send is admitted, show Reply unconfirmed: only the ACK may be missing. Before Send, use Request expired. No Retry or late ACK reopens the question. Explicit restart / Data retained Write 'restart Cody with code 043533477' in the current message: all nine digits, including the zero. This prevents misunderstandings; it is not a password and must not be completed from documents. Use the paired LAN peer; retain data. ACK means accepted; boot needs fresh presence. Factory reset / First setup 'Erase Cody settings with code 043533477' removes preferences, Wi-Fi and pairing while preserving firmware, assets and SD. ACK means admission. Restart only after verified erasure, then resume Wi-Fi setup. Errors stay visible and never trigger reboot.
10 October 2026 ERRATA	**Errata and corrections** Affected: Original planning page 46; original screen IDs retained in the description. Original pages 1, 2, 11, 12, 13, 14, 16, 21, 22 Affected main pages are updated; unrelated pages and previous addenda are preserved. References use screen IDs. Screens 21-27 and 31-32 A03 is applied: PRs stay read-only and 25 is retired. View log and the old vertical list are removed. Screen 27 now has cards and a Send summary; 31 requires delivered after the recipient's ACK. Initial analysis gap Receipt, delivery and expiry were ambiguous, with outdated PR illustrations. Separate them using tickets, received/delivered and 180 seconds. Timeout and offline never choose answers. History, commands and evidence Restart and factory reset are separate: restart retains data; factory reset clears configuration and returns to setup. ACK means admission. Preserve history and real errors; host and physical results are in done.md.

Revision register

Historical dates, original screen references and evidence are retained in this final grid.

DATE / TYPE	REVISION DETAILS AND ORIGINAL REFERENCES
10 October 2026 CHANGE	**A single final revision register** Affected: Previous pages 47-50; earlier history on pages 13-46 (Navigation) or 23-46 (Planning). The 10 October screen supplements previously on pages 47-50 are reorganized into this grid. Their dates, decisions, screen IDs and evidence limits remain explicit. All preceding 46 pages, including historical artwork and earlier corrections, remain intact. Earlier decisions are historical: 03 October A01 separates installation and provisioning; 04 October A02 preserves the skin selector's caller, A03 keeps PR status read-only, and A04 gates pairing handoff. The 05-06 October idle and branding records, and the 09-10 October setup, retained pairing, activity and notification revisions remain in their preserved pages. Evidence / limits: Document reorganization only. Original physical page numbers remain valid through page 46; later references use stable screen IDs.
10 October 2026 ADDENDUM	**Boot and Wi-Fi** Affected: Original page 47; branding, screens 01-09, 39, 40 and 42. Cody AI Companion and A distraction-free sidekick appear together above the complete floppy artwork. Its 42 frames retain the 3.5-second cycle; startup work can extend wall time. Title y=60, subtitle y=110, artwork (137,164), 192 x 219. After local startup, absent Wi-Fi opens Welcome 04, missing association/IP opens 42, a valid LAN without pairing opens 40, and retained pairing searches its PC on 14. Screen 05 opens the installation QR on 06 without starting radios; Pair Cody starts Bluetooth on 07. The session QR exists only at ReadyToScan and retires when credentials or connection are in progress. Cancel waits for confirmed cleanup. Screen 09 admits Continue only with current valid Wi-Fi/IP. Screen 42 offers Network setup or Later. Local LAN and public Internet are separate facts. Debug handoff 39 requires a fresh tap; Release continues to 09.

Revision register

Historical dates, original screen references and evidence are retained in this final grid.

DATE / TYPE	REVISION DETAILS AND ORIGINAL REFERENCES
10 October 2026 ADDENDUM	**Pairing and connection** Affected: Original page 48; screens 12-15, 40 and 41. Hold the visible character still for 1.5 seconds to open Setup. Connect Codex and Network setup share a row; Options is centered below. Release the opening touch before selecting. Back follows the valid previous route. Already paired offers Keep pairing, Pair again and Forget. Only Pair again authorizes replacement; Forget needs a second confirmation. Pairing paused identifies an incomplete candidate. Pairing error preserves the storage cause and permits only explicit removal. Preparing secure code / Please wait / About 30 seconds is an estimate, not measured progress. Use all six characters on the intended Cody after discovery. The backend pairing window is five minutes; screen inactivity is three minutes from code readiness. You're all set follows the two-sided persisted handshake. Let's go searches the retained PC immediately, without the earlier ten-second Idle detour. Security clarification: a signed READY now challenges the connection; the broker admits presence and releases current data only after the fresh endpoint-bound COMMITTED response. Genuine prompt hooks establish chat activity, independently of pairing or presence.
10 October 2026 ADDENDUM	**Settings and complete controls** Affected: Original page 49; screens 10, 11, 36 and 41. Shake to wake, Auto-rotate and Sounds retain persistent On/Off values. Orientation shows Auto when auto-rotation is on; manual orientation is 0 or 180 when off. Sounds Off stops playback while preserving microphone and wake-word operation. Next opens Settings / Personalization / Choose Cody's look. Continue opens the stable-ID skin selector: Pearl Mist, Aurora Veil, Cosmic Stardust, Enchanted Mosaic and Dreamwave. A save preserves the prior value until successful write/readback. Save failed and Retry retain the cause and requested value; no silent repair is admitted. Two actions use side-by-side gradient cards; three add one centered card below. A single action uses an icon-left capsule. Captions never use teletype: controls and touch become available together after title and body narration. Preserve QR quiet zones and the local circular chord.
10 October 2026 ERRATA	**Current screen corrections and evidence** Affected: Original page 50; current base pages 4, 5, 6, 7, 8; branding, screens 01-15, 36, 40-42. Replace the earlier title-free/twofold boot drawing with the current title, subtitle and 192 x 219 artwork. Replace obsolete setup, secure-code preparation, connection detour and settings descriptions with the current contracts above. Historical values and diagrams retain their dated meaning. The source-based static schematics use illustrative fonts and values. They are neither simulator frames nor device captures. Layout coordinates are not sensor measurements. This PDF revision does not verify firmware, simulator, hooks or physical behavior. Evidence / limits: English grid replaces the latest four appendix pages; prior corrections and unrelated pages are preserved.

Revision register

Historical dates, original screen references and evidence are retained in this final grid.

DATE / TYPE	REVISION DETAILS AND ORIGINAL REFERENCES
11 October 2026 ERRATA	**English throughout the document** Affected: Original pages 1,2,9-14,16,21,22,39-46; former revision-register page 53. Historical Italian headings, captions and body copy were translated into English while preserving dates, original page and screen IDs, technical values and evidence limits. The original English correction on former page 53 is consolidated here. The connection atlas, four-page A5 flyer and manga status atlas required no language correction in that revision. Evidence / limits: Document language and visual review only; no firmware or physical verification.
11 October 2026 CHANGE	**Restore the illustrated review workbook** Affected: Original main pages 1,2,5-16,21,22; screens 21-35B and 39-42; restored voice pages 23-24. Restore the cyan Cody artwork and separate writable NOTES areas from the 02 October source 027498aa. Preserve layout geometry, expression studies, debug and all 256 palette slots. Replace white placeholder screens with coherent illustrated current routes. Standalone appendices from original pages 23-46 and the 47-53 revision pages are consolidated into this single final grid. Historical dates, decision substance, original page/screen references and evidence are retained. Security remains a clearly identified subsection of this register. Screens 28-30 restore the one-way voice design: wake word, double tap or button starts a dedicated audio route; suspend character rendering, use a precomputed animation, stream over authenticated encrypted Wi-Fi and retain up to ten seconds of unacknowledged audio in PSRAM. Native Codex audio input remains research, not implemented firmware. No OTA or audio implementation is claimed. Screens 33/34 now use short manga notices without percentages: low at 20 and 10 percent, charging entered once, full only on fresh measured 100 percent. Admit after ten seconds of uninterrupted free Home; never interrupt a request or prolong standby. Continuing charging remains silent. Evidence / limits: PDF authoring and visual review. Source-based battery changes are tracked separately; no build, simulator or physical validation.

Revision register / Security

Historical dates, original screen references and evidence are retained in this final grid.

DATE / TYPE	REVISION DETAILS AND ORIGINAL REFERENCES
10 October 2026 SECURITY	**Fresh proof before presence or current data** Affected: Screens 12-15, 18U, 26-27, 31-32; CCP FIND / READY / COMMITTED. A captured authenticated broadcast FIND must not establish broker presence, select a notification destination or obtain current ACTIVITY, TECH or EVENT data. The broker issues a random challenge, signed together with the observed device IPv4 address and UDP port. Cody verifies that endpoint before answering. READY and COMMITTED use explicit ready-v2 and committed-v2 authentication domains and the fields device, pc, nonce, challenge, address, port and proof. Only the matching one-time challenge response from the challenged endpoint admits liveness, commit recovery and current-data delivery. Old and new firmware/plugin implementations must be updated together; no legacy handshake fallback is provided. Evidence / limits: Source implementation and static review. CCP 0.6.3 is not deployed or physically verified by this document revision.
10 October 2026 SECURITY	**Bound unauthenticated work** Affected: Firmware UDP worker; PC discovery and pairing; CCP ingress and runtime logs. Firmware limits the receive pass by packet count and elapsed time so local commands, expiry and protocol maintenance still run while UDP remains busy. Fixed-size global/per-source ingress admission precedes expensive authentication; source-address rotation cannot remove the global budget. PC discovery bounds total offers and per-source contributions, and pairing bounds distinct verification work and time. Overflow is an explicit failure, rather than a partial result presented as complete. The broker applies ingress limits before expensive credential access and aggregates rejection diagnostics; rotating logs bound retained storage. Initial limits: firmware receives at most eight packets or a 5-ms slice, with 32 packets/s globally (burst 64), 16/s per IP (burst 32), and 16 source slots. PC UDP allows 64/s globally (burst 128), 16/s per IP (burst 32), and 64 slots. Discovery admits 16 offers, four per IP, and 128 received packets; pairing verification has a two-second deadline before candidate persistence. Endpoint challenges have a two-second lifetime and a 128-entry cap. Rejection summaries appear at most once per five seconds. Broker logs retain at most three 2-MiB files; local control admits at most 32 concurrent workers, retaining its three-second socket deadline and 64-KiB request-body limit. Evidence / limits: Initial tuning constants require measurement against real pairing/reconnection bursts. No flood probe, suite, build or flash is claimed.
10 October 2026 SECURITY	**Retained safeguards and deferred protocol work** Affected: Pairing code and link-key derivation; ACTIVITY / TECH / EVENT confidentiality; optional device HTTP. Keep local user admission, the expiring pairing window, constant-time HMAC verification, separate active/pending durable records, strict identities and packet bounds, authenticated choices/settings/debug/reset commands, and loopback-only authenticated PC control. Native approval remains on the computer. Failed storage or authentication must not become success. The six-character code still permits offline candidate checking from a captured pairing transcript; online rate limiting does not remove that risk. CCP payloads remain authenticated but unencrypted. Evaluated PAKE or high-entropy QR pairing, followed by standard authenticated encryption and replay discipline, is deferred as a coordinated protocol redesign. No custom cryptography or confidentiality claim is introduced. Optional device HTTP behavior is a separate security scope: enabling its plaintext bearer API or broad health endpoint requires its own review. This CCP hardening does not claim to secure an enabled device HTTP service or a hostile local machine. Log scope: broker/network diagnostics rotate under the exclusive broker owner. The startup-only runtime.log still appends across launches; LAN UDP rejection traffic cannot write to it. Bounded multi-process bootstrap logging remains a separate follow-up. Evidence / limits: Remaining risks are recorded in planned-todo.md. Compilation, device stack headroom and physical pairing/recovery checks remain pending in TODO.md.